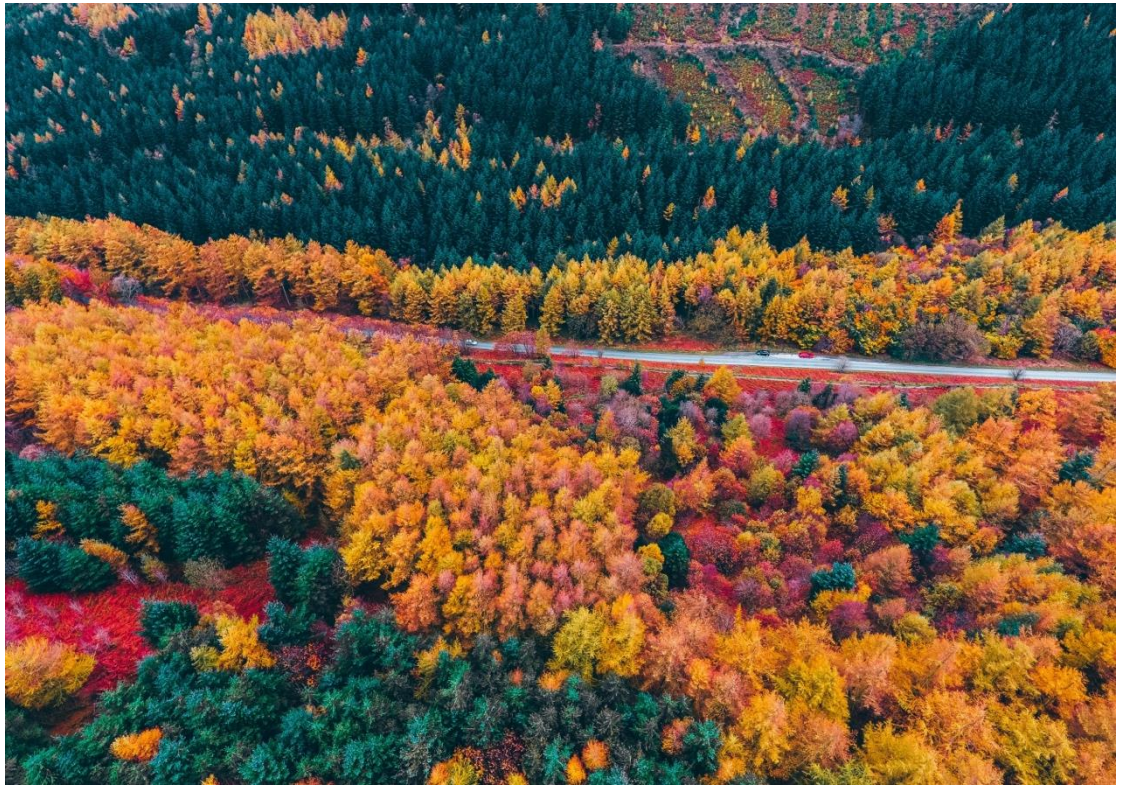
GEOG 172

Global Statistics

Well, getting ready for clustering statistics

Lecture 6-1





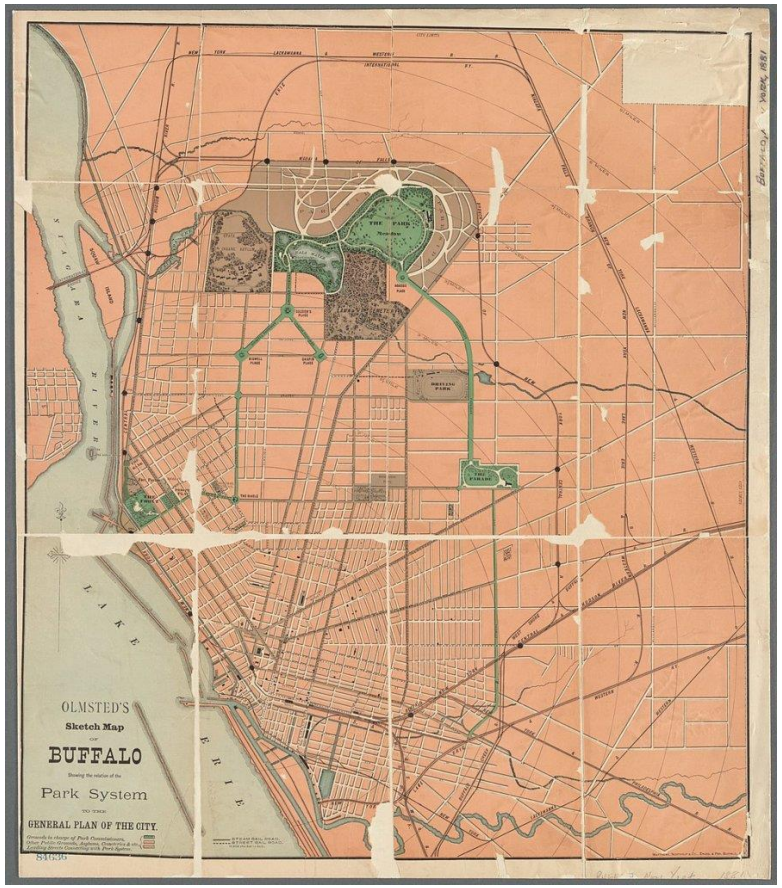
NATURAL AREAS

Entities modeled using boundaries defined by natural phenomena



ANTRHROPIC AREAS







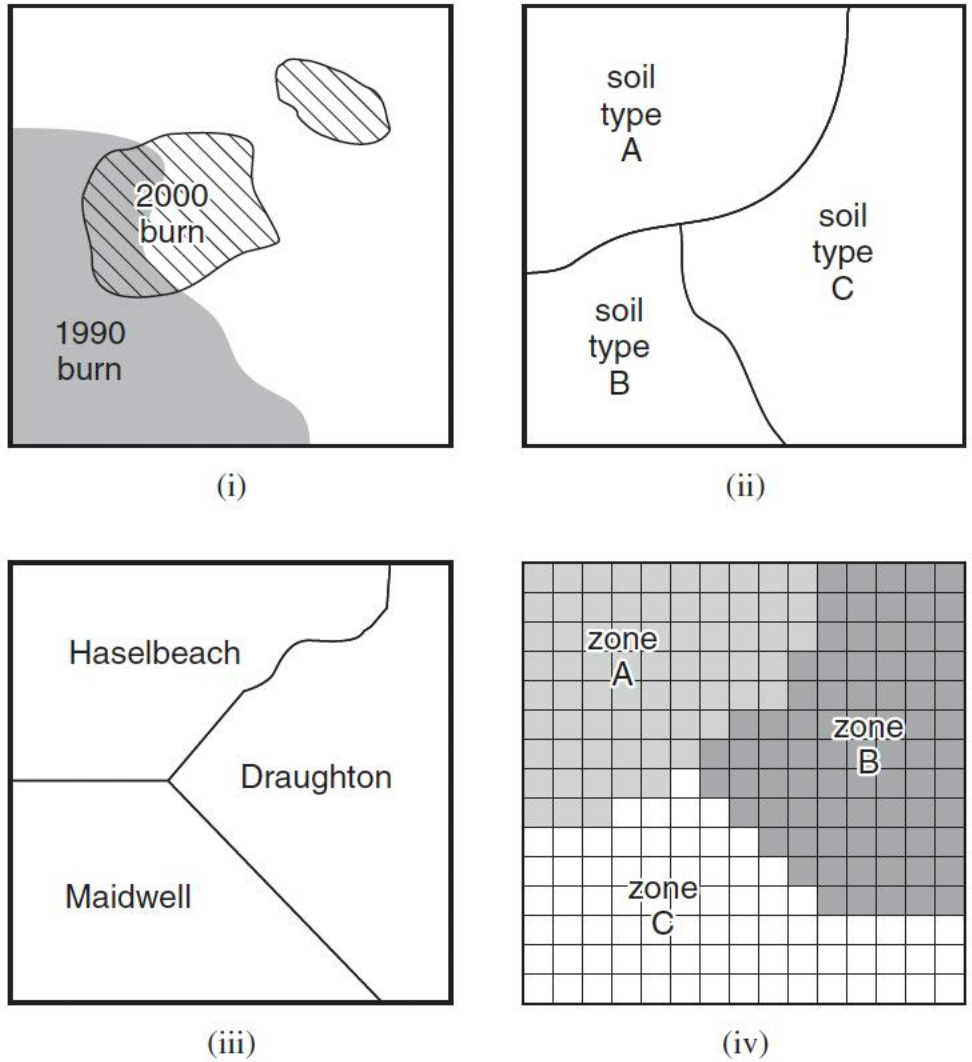


Figure 7.1 Types of area objects: (i) a pattern of forest burns over natural areas that overlap and are not planar enforced; (ii) natural area objects resulting from an interpreted mapping, in this case soil type; (iii) imposed areas, in this case three civil parishes; and (iv) an imposed raster recording three types of soil.

QUESTION

What are some of the ways we can measure spatial relationships?

QUESTION

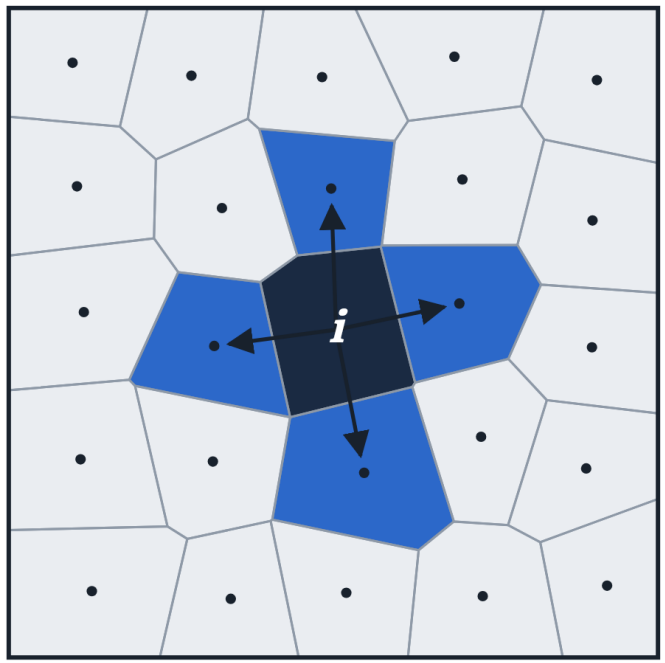
What are some of the ways we can measure spatial relationships?

ANSWER: We can measure distance, but this can include – Euclidean, Great Circle, Manhattan, Travel Time, … Distance can also be measured using adjacency, but this can also be defined in many ways. Interaction is another possibility.

Spatial Weights

From concept to statistical representation

1.1 k-Nearest-Neighbor BINARY



Focal unit i and its $k = 4$ nearest neighbors by centroid distance, each with $w_{ij} = 1$.

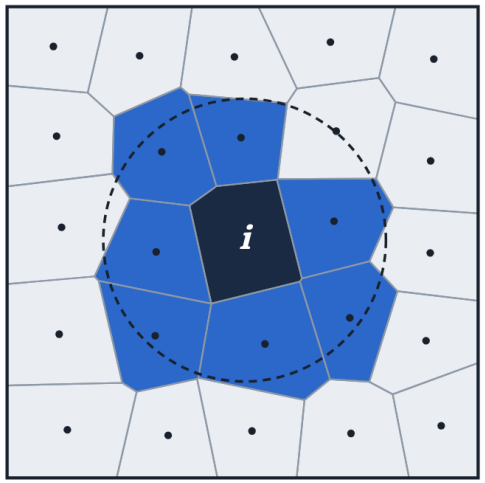
FORMULA

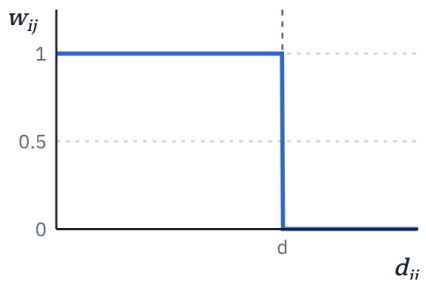
$$w_{ij} = \begin{cases} 1, & j \in N_k(i) \\ 0, & \text{otherwise} \end{cases}$$

DESCRIPTION

Rank every other unit by its centroid distance from i. The k closest units get weight 1 and all others get 0, so all k neighbors count equally. The weights are usually asymmetric: j can be one of i's nearest neighbors without i being one of j's.

1.3 Radial Distance BINARY





Units whose centroids lie inside the circle of radius d around c_i get $w_{ij} = 1$. The curve shows the step at d.

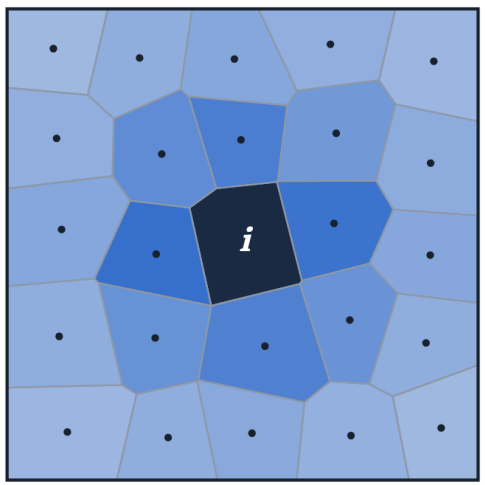
FORMULA

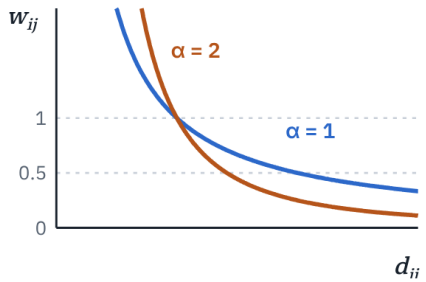
$$w_{ij} = \begin{cases} 1, & 0 \leq d_{ij} \leq d \\ 0, & d_{ij} > d \end{cases}$$

DESCRIPTION

Every unit within a threshold distance, or bandwidth, d of i gets weight 1, and units farther away get 0. Influence does not decline inside d. Useful when a natural cutoff exists, such as walking distance or a one-hour drive.

1.4 Power-Distance NON-BINARY





Darker shading means a larger w_{ij} (map: $\alpha = 1$). The curve compares $\alpha = 1$ and $\alpha = 2$.

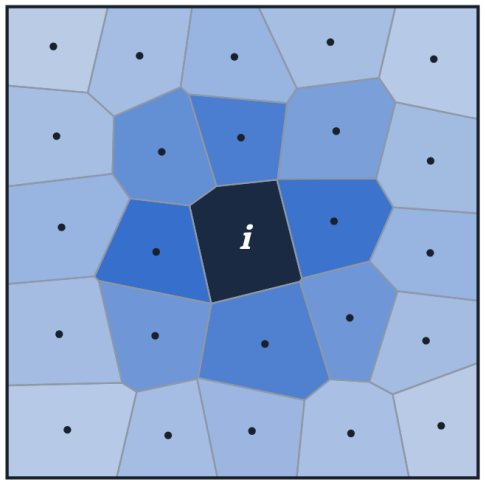
FORMULA

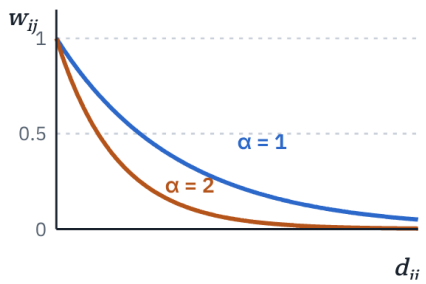
$$w_{ij} = d_{ij}^{-\alpha}$$

DESCRIPTION

Influence falls continuously with distance as a negative power function. A larger α means a faster decline; $\alpha = 1$ or 2 is typical. This is the same function used in inverse distance weighting (IDW).

1.5 Exponential-Distance NON-BINARY





Darker shading means a larger w_{ij} (map: $\alpha = 5$ per map unit). The curve compares $\alpha = 1$ and $\alpha = 2$.

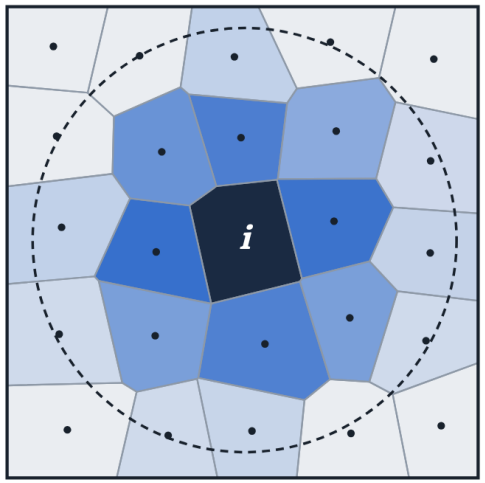
FORMULA

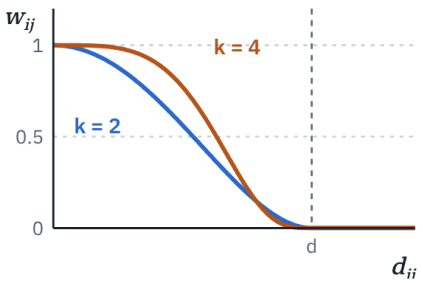
$$w_{ij} = \exp(-\alpha\, d_{ij})$$

DESCRIPTION

Influence falls as a negative exponential of distance. Unlike the power function, it stays finite as $d_{ij} \to 0$, but it drops toward zero faster at long distances.

1.6 Double-Power-Distance NON-BINARY





Shading shows w_{ij} with bandwidth d (circle) and $k = 2$. The curve compares $k = 2$ and $k = 4$.

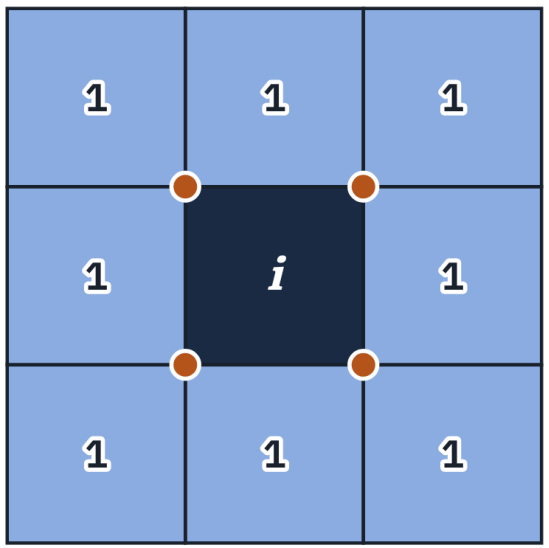
FORMULA

$$w_{ij} = \begin{cases} \left[1 - (d_{ij}/d)^k\right]^k, & 0 \le d_{ij} \le d \\ 0, & d_{ij} > d \end{cases}$$

DESCRIPTION

A smooth, bell-shaped decline that reaches exactly zero at the bandwidth d and stays zero beyond it. Typical values are $k = 2, 3, 4$; $k = 2$ gives the quadratic distance function.

2.1 Queen Contiguity BINARY



All 8 surrounding cells are neighbors, including the 4 that touch i only at a corner (dots).

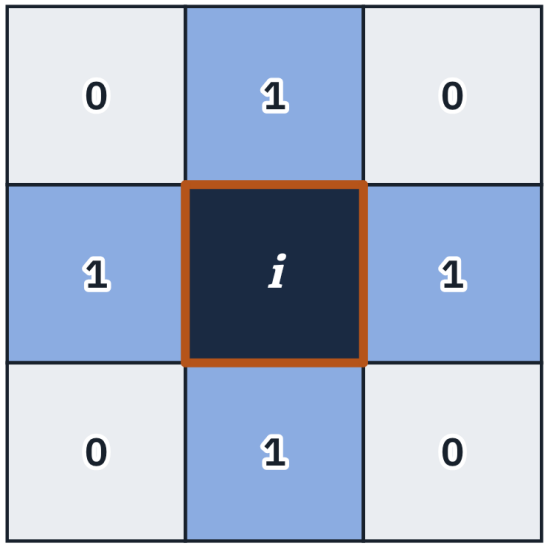
FORMULA

$$w_{ij} = \begin{cases} 1, & \text{bnd}(i) \cap \text{bnd}(j) \neq \emptyset \\ 0, & \text{bnd}(i) \cap \text{bnd}(j) = \emptyset \end{cases}$$

DESCRIPTION

Two units are neighbors if their boundaries touch at all, even at a single point, just as a queen can move diagonally through a corner.

2.2 Rook Contiguity BINARY



Only the 4 cells that share an edge (bold) with i are neighbors.

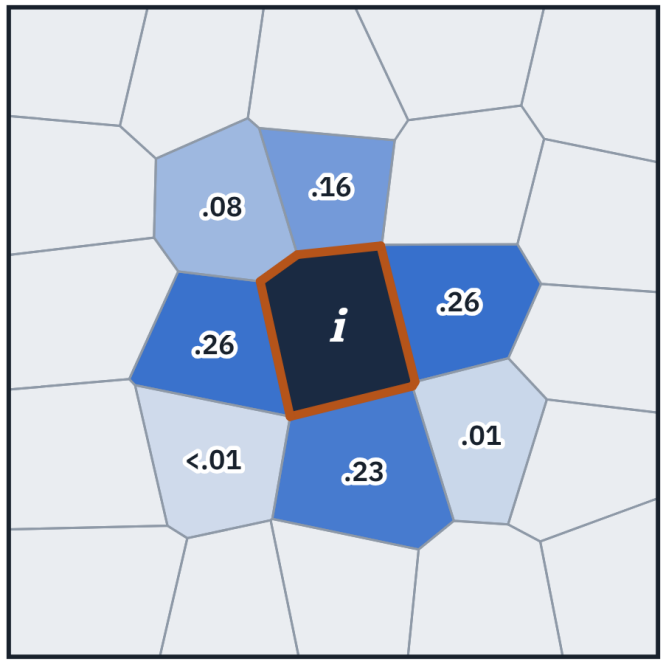
FORMULA

$$w_{ij} = \begin{cases} 1, & l_{ij} > 0 \\ 0, & l_{ij} = 0 \end{cases}$$

DESCRIPTION

A stricter rule: neighbors must share a boundary segment of positive length. Units that meet only at a corner are excluded, just as a rook moves only through edges.

2.3 Shared-Boundary NON-BINARY



Bold lines mark i's shared boundary. Each number is that neighbor's share l_{ij}/l_i; together they sum to 1.

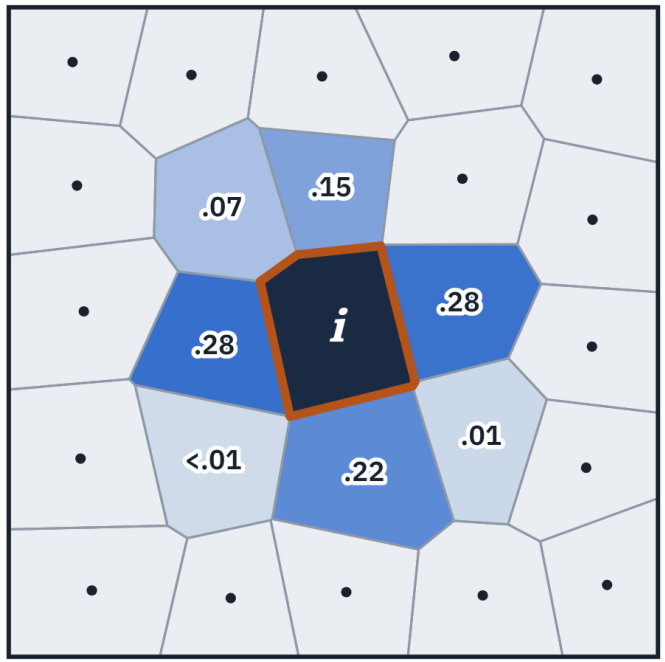
FORMULA

$$w_{ij} = \frac{l_{ij}}{l_i} = \frac{l_{ij}}{\sum_{k \neq i} l_{ik}}$$

DESCRIPTION

The weight is the fraction of i's total shared boundary that it shares with j, so each row sums to 1 by construction. The matrix is generally not symmetric: $l_{ij} = l_{ji}$, but l_i and l_j usually differ.

3.1 Combined Distance-Boundary NON-BINARY



Numbers give w_{ij} with $\alpha = 1$. A long shared border and a short centroid distance both raise the weight.

FORMULA

$$w_{ij} = \frac{l_{ij}\, d_{ij}^{-\alpha}}{\sum_{k \neq i} l_{ik}\, d_{ik}^{-\alpha}}$$

DESCRIPTION

Combines shared-boundary length with power-distance decay, then row-normalizes. Cliff and Ord (1969) found this the best scheme for the Eire blood-group data, using $\alpha = 1$.

Spatial Speed Networking

Finding your spatial vibe

ACTIVITY

Overview:

You are about to participate in a spatial networking event. This is a LinkedIn-style professional networking event, with your own profile of a unique geographic problem. You need to connect with others who have a completely different looking profile, different study object, but perhaps the same way of framing the spatial relationship or conducting distance measurements between study objects.

Purpose:

Identify the **spatial objects, process, and meaningful spatial relationships** embedded in a real-world geographic problem

ACTIVITY

Instructions:

Turn your case into a "Spatialin Profile"

- what geographic **objects** are being studied;
- what **process** is of interest;
- what could make two objects spatially related;
- how that relationship could be measured using;
- which spatial weights scheme best represents that relationship

ACTIVITY

Distance Skill:

- Adjacency, Euclidean, Great-circle, Manhattan, Network, Travel-time

Spatial Weights Skill:

- Queen, Rook, Shared Boundary, K-Nearest Neighbor, Radial Distance, Power Distance, Exponential Distance,

ACTIVITY

Example:

Wildfires can spread from one forest management unit into another. We are interested in identifying which management units should be treated as directly connected when studying the spread of fire.

Spatial Object?

Spatial Process?

Distance Skill?

Weights Skill?

ACTIVITY

Blue Sticker: - We see spatial relationships the same way

Red Sticker: - We share the same spatial weights skill

Green Sticker: - Top professional match

TAKEAWAYS

- Different type of areal objects exist and can be related to one another in space using different types of distances
- The weights matrix is the most common way to represent spatial relationships, and many options exist
- A goal is to use weights that match what we think the important relationships driving the spatial process are